

Design and Optimisation of a Two-stage Amplifier Based on a Differential Input Stage and a Common-source Amplifier

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Abstract. Since the emergence of integrated circuits, through the circuit structure, the production process of continuous iterative updating, so that it's used in various types of electronic equipment more and more widely. Currently, operational amplifier circuits play an extremely important role in signal processing and communication systems. Amplifiers have different characteristics depending on the needs of different electronic devices. This thesis addresses the optimisation and design of the characteristics of the two-stages amplifier. Firstly, several classical amplifier structures are analysed. Select the appropriate structure and combine it into a two-stages amplifier. Simulation is carried out on cadence software and later the simulation results are analysed before parameter optimisation. On this basis, a two-stages amplifier based on a PMOS differential input stage with a common-source amplifier has been designed. It also improves the amplifier's gain, frequency range, gain bandwidth, and other performance metrics. Using PMOS self-bias circuit to improve power supply stability. Finally, based on the development hotspots of integrated circuits, the possibility of circuit optimization was analyzed. Discussions were conducted on negative feedback circuits, phase compensation methods, and other aspects.

Keywords: Two-stage amplifier; differential input stage; common source amplifier; cadence.

1. Introduction

With the development of modern electronics, the performance of the amplifier directly affects the overall effectiveness of the system. There has been a gradual shift in the design of operational amplifiers towards customised types. Design an amplifier with excellent performance while balancing all performance parameters. Amplifier gain can be increased by using a common-source, common-gate structure in the output stage, but the common-source, common-gate structure can severely limit the output voltage swing as the supply voltage decreases [1].

On the other hand, op-amp circuits are closely related to the degree of development of CMOS processes. From the familiar mp3, and mp4, to the present PDA, watch, and so on, these electronic products require very high speed and precision of the circuits [2]. As a result, the requirements for the devices are getting higher and higher. In turn, more circuit designs are produced that meet the requirements. In 2005, National Semiconductor Corporation successfully completed two high-speed, low-distortion, fully differential op-amps (LMH6550 and LMH6551) using a fully differential voltage feedback circuit design with a +5V or -5V supply voltage. The advantages are a bandwidth of up to 400 MHz and a spurious signal-free dynamic range of - 86 dBc [3]. as the process feature size

decreases, the DC gain of the amplifier decreases dramatically, resulting in a serious degradation of the amplifier's performance [4]. In order to meet the overall system requirements, two-stage or multi-stage operational amplifiers have become a hot research topic.

In terms of integration, it ranges from small-scale (SSI) to medium-scale (MSI) and large-scale (LSI). Up to the current very large scale (VLSI) and ultra large-scale integrated circuits (ULSI). It can be seen that integrated circuits have undergone tremendous changes in recent decades [5].

This thesis presents a two-stage operational amplifier. A PMOS differential input, a typical common source stage structure, a PMOS self-bias circuit, and a Miller frequency compensation circuit are used to improve the performance of the operational amplifier. To simulate and analyse the two-stages operational amplifier circuit in cadence environment for each performance index and the parameters were optimised.

2. Fundamentals and Theory of Two-Stage Amplifier Design

2.1. Circuit structure and basic principle

In this design, the first stage amplifier is a differential amplifier, which can effectively reduce the effect of noise and reduce the input misalignment. The OTA five-tube structure is used, which is more suitable for IC integration by changing the bias current of the OTA, thus changing its interconductance gain g_m and voltage amplifier gain A_u [6]. The second stage amplifier is a common-source amplifier capable of providing a large gain with a large input impedance. For the design of the biasing circuit, a PMOS self-biasing circuit is chosen in this design, where the output current is weakly correlated with the power supply V_{CC} . Finally, there is also a phase compensation circuit. Each additional amplification stage will generate at least one high impedance point. This in turn will introduce a pole, Excessive poles can greatly affect the stability of the amplifier [7]. The phase compensation circuit ensures the stability of the circuit.

2.2. Circuit analysis

2.2.1. Bias circuit. consisting of three pairs of current mirrors and keeping the left and right currents the same. Consistent aspect ratio for each pair of mos tubes. The magnitude of the current is related to $M5M6$ and $R0$. Because the V_{GS} of the $M5M6$ is different, the V_{GS} of the $M5$ will be smaller due to the presence of the resistor. So the shape aspect ratio needs to be adjusted to keep the currents on both sides similar. The voltage difference because of the difference in V_{GS} and the magnitude of $R0$ determines the magnitude of the current. A bias voltage is supplied to the $M14$ gate along with the bias current, allowing the phase compensation circuit to operate in the proper interval. As shown in fig 1.

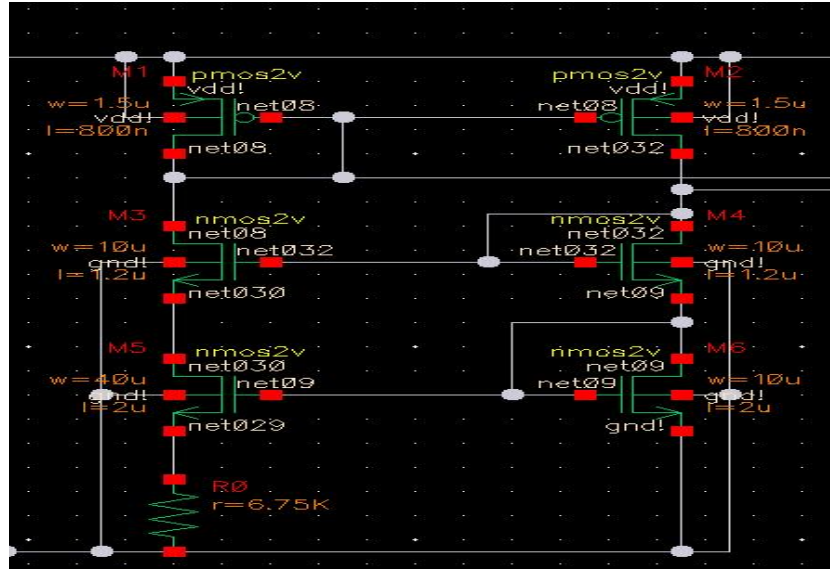


Fig 1. Bias circuit diagram

2.2.2. *PMOS differential amplifier.* M7M8 is used as a differential input pair, and the aspect ratio needs to be consistent to eliminate interference from noise and other factors. The M3M4 should have exactly the same aspect ratio as an NMOS current mirror load tube. M12 acts as a PMOS current mirror and as a tail current in the first stage of the amplifier circuit, providing a stable bias. Current mirror loaded structure means, the load of the differential amplifier is replaced by a current mirror[8]. Ensure that the circuit can work stably in the amplification region while reducing noise and distortion in the circuit. As shown in fig 2.

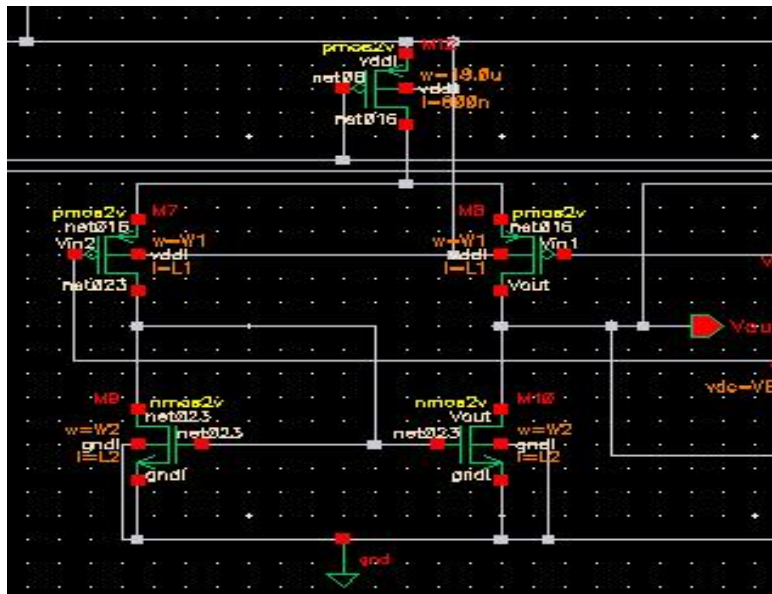


Fig 2. Differential circuit diagram

2.2.3. *Common source level amplifier circuit and phase compensation circuit.* M13 acts as a PMOS current mirror and acts as a load for the second stage amplifier, where the current changes through the load thus generating an output voltage signal. Make M14 operate in the linear region so that it is equivalent to a resistor. By adjusting the size of the resistor, the action of moving the zero point of the

right half plane can be achieved. The mos tube is connected across the input and output of the second stage along with the capacitor C0 to form the RC Miller compensation. The role is to ensure a suitable phase margin. As shown in fig 3.

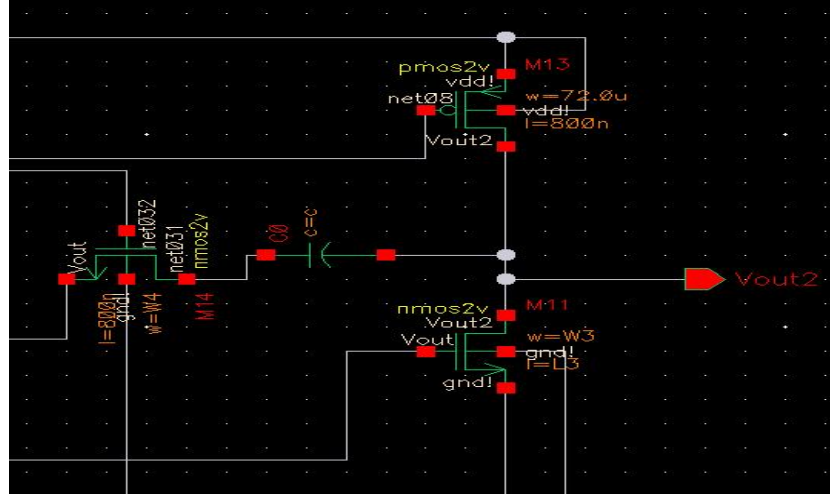


Fig 3. Common source level stage and phase compensation circuit diagrams

3. Analysis and Optimisation of Cadence Simulation Results

3.1. Circuit structure and basic principle

During the commissioning of the first stage, it is necessary to make the gain as large as possible while the mos is in the saturation region. Therefore, the choice was made to change the aspect ratio of the mos for debugging. Secondly, in order to match the primary and secondary amplifiers, it is required that $VGST9=VGST10=VGST11$, $IDS9/(W/L)9=IDS10/(W/L)10=IDS11/(W/L)11$. The amplification gain interval is obtained after satisfying the above conditions. As shown in fig4. The red line is the primary gain = 100 and the total circuit gain = 1.3k.

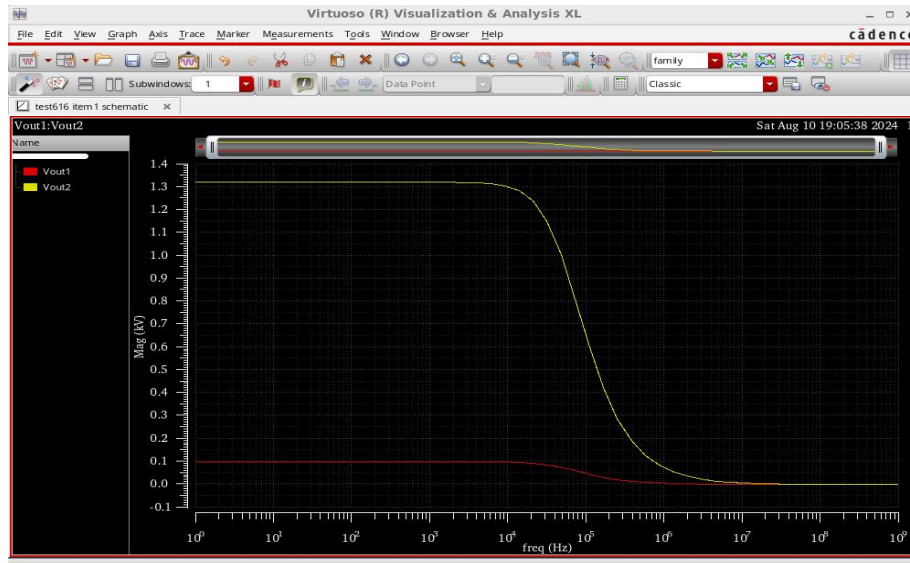


Fig 4. Circuit gain diagram

In phase compensation circuits, C0 enlargement enhances the pole-splitting function, reduces the input integration noise, reduces the second stage power consumption, and improves the phase margin.

However, the disadvantage is that it reduces the GBW and the slew rate. The preset C0 is thus changed to 1.2 pf. Next, make sure that the NMOS is in the linear region. The phase margin and op-amp gain bandwidth are shown in fig5. From the red line, the GBW of the op-amp is 67 MHz with a phase margin of 70°.

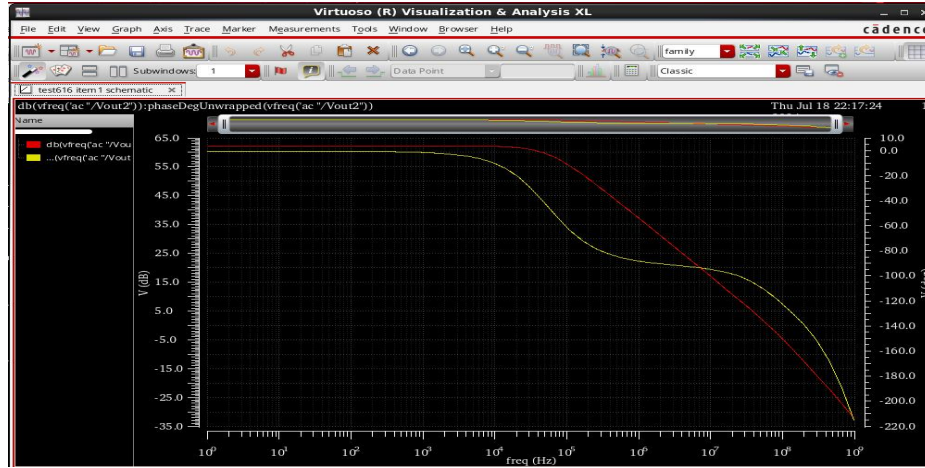


Fig 5. Gain bandwidth and phase margin chart

3.2. Circuit structure and basic principle

By introducing unit gain negative feedback in the secondary amplification region, as shown in fig 6. Connect the output to the negative terminal of the input, at which point V_{out} is no longer determined by the parameters of the load and amplifier tubes. Because the DC gains of the secondary op-amps are all high enough, V_{in+} approximates V_{in-} . In summary, V_{out} is directly determined by the DC voltage of V_{in+} . Under the premise of negative feedback, the op-amp can automatically adjust the V_{gs} of the amplifier so that the amplifier current is constantly equal to the saturation current of the load tube. Thus the boundary condition that guarantees simultaneous saturation of the load and amplifier tubes becomes a range.

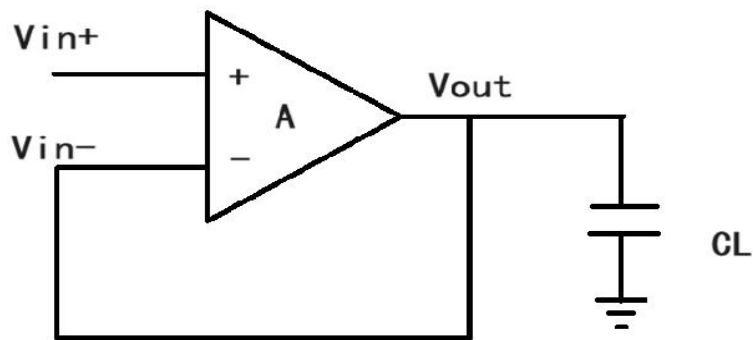


Fig 6. Unit gain negative feedback circuit diagram

In terms of power consumption, in 2018, the Belgian Microelectronics Research Centre launched 3nm chips, and lower-size chips are also under research, so the design of semiconductor integrated circuits with low voltage and low power consumption has become one of the research hotspots in the center of the international research institutes [9]. Many new compensation methods have been proposed for phase compensation. Damping Factor Controlled Frequency Compensation (DFCFC)

greatly improves the GBW by adding a damping network. However, DFCFC is suitable for large load capacitance. For smaller load capacitances, DFCFC does not show good characteristics [10].

4. Conclusion

The present design of the two-stage amplifier achieves a large gain multiplier over a large frequency range by combining different characteristic amplifier circuits and adjusting the component parameters. Future developments for two-stage or multi-stage operational amplifier circuits aim to improve the circuit or introduce negative feedback to improve the performance of the circuit. It can be seen that the performance of operational amplification circuits in the enhancement of diversification, can be customised through the specific needs of targeted amplification circuits, with the improvement of the overall performance of the equipment, which is also a popular direction for the future development of operational amplification circuits. On the other hand, while cascading different amplifiers, studying negative feedback circuits is also one of the research hotspots. Introducing high-performance negative feedback can effectively enhance the stability of the circuit. Ensure stable amplification under external environmental interference.

References

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